

## Kitchen\_4

Level: Level 0 · Internal condition: 1 Bed Apt. Kitchen - Kitchen\_4

## DESIGN CRITERIA

|                            | Heating | Cooling |
|----------------------------|---------|---------|
| Room set point             | none    | none    |
| Outdoor dry bulb           | —       | —       |
| Outdoor RH (coincident)    | —       | —       |
| <b>ROOM HUMIDITY</b>       |         |         |
| Humidification set point   | —       |         |
| lower RH limit             |         |         |
| Dehumidification set point | —       |         |
| upper RH limit             |         |         |

## SIZING (TAS DESIGN LOADS)

|                      | Heating              | Cooling              |
|----------------------|----------------------|----------------------|
| Design load          | 0 W                  | 0 W                  |
| Design load per area | 0.0 W/m <sup>2</sup> | 0.0 W/m <sup>2</sup> |
| Sizing multiplier    | not set              | not set              |

Design loads: from Tas sizing — date/currency not recorded

## HEATING

| Design day          | Leeds_TRY ANN HTG 100% CONDS DB |                           |
|---------------------|---------------------------------|---------------------------|
|                     | Design day                      | Full year                 |
| Peak sensible load  | 1,706 W                         | 911 W                     |
| Peak hour           | 13:00–14:00                     | 1 Jan 00:00–01:00 (HOY 1) |
| Room dry bulb       | 14.4 °C                         | 13.6 °C                   |
| Room resultant      | 13.0 °C                         | 12.8 °C                   |
| Room RH             | 22 %                            | 58 %                      |
| Room humidity ratio | 2.2 g/kg                        | 5.6 g/kg                  |
| Outdoor dry bulb    | —                               | 3.3 °C                    |
| Outdoor RH          | —                               | 100 %                     |

## SENSIBLE LOAD COMPONENTS AT PEAK

| Component                     | Design day (W) | Full year (W) |
|-------------------------------|----------------|---------------|
| Equipment (sensible)          | 0              | 50            |
| Infiltration / ventilation    | -311           | -157          |
| Building heat transfer        | -686           | -478          |
| External conduction — opaque  | -481           | -215          |
| External conduction — glazing | -228           | -111          |

Zero at both peaks (not listed): Solar, Lighting, Occupancy (sensible), Air movement (inter-zone), Air handling unit. Latent gains: zero at both peaks (Occupancy, Equipment).

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

## COOLING

| Design day          | Leeds_TRY ANN CLG 0% CONDS DB=>GRad |                               |
|---------------------|-------------------------------------|-------------------------------|
|                     | Design day                          | Full year                     |
| Peak sensible load  | 1,288 W                             | 863 W                         |
| Peak hour           | 00:00–01:00                         | 19 Jul 19:00–20:00 (HOY 4796) |
| Room dry bulb       | 20.1 °C                             | 19.1 °C                       |
| Room resultant      | 21.3 °C                             | 19.9 °C                       |
| Room RH             | 88 %                                | 58 %                          |
| Room humidity ratio | 13.0 g/kg                           | 7.9 g/kg                      |
| Outdoor dry bulb    | —                                   | 15.4 °C                       |
| Outdoor RH          | —                                   | 80 %                          |

SENSIBLE LOAD COMPONENTS AT PEAK

| Component                     | Design day (W) | Full year (W) |
|-------------------------------|----------------|---------------|
| Solar                         | 0              | 61            |
| Lighting                      | 0              | 150           |
| Occupancy (sensible)          | 0              | 19            |
| Equipment (sensible)          | 50             | 300           |
| Infiltration / ventilation    | -13            | -137          |
| Building heat transfer        | 851            | 374           |
| External conduction — opaque  | 442            | 153           |
| External conduction — glazing | -42            | -56           |

Zero at both peaks (not listed): Air movement (inter-zone), Air handling unit

LATENT GAINS AT THE SENSIBLE PEAK HOUR

| Component          | Design day (W) | Full year (W) |
|--------------------|----------------|---------------|
| Occupancy (latent) | 0              | 14            |

Zero at both peaks (not listed): Equipment (latent)

Latent gains: internal gains only (occupancy, equipment) at the hour of the sensible peak. The room's latent load, which also includes moisture from infiltration and ventilation, is not in the results and is not derived.

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

|                       | Heating            | Cooling            |
|-----------------------|--------------------|--------------------|
| Status                | Available          | Available          |
| Result source         | SAM.Analytical.Tas | SAM.Analytical.Tas |
| Read into model       | 27 Sep 2026 20:42  | 27 Sep 2026 20:42  |
| Matches current model | not recorded       | not recorded       |

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year, and its hour of the year (HOY 1 = 1 Jan 00:00–01:00, HOY 8760 = 31 Dec 23:00–24:00).

Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.